

Self-Rambling: A Stealthy Inference Cost Attack in Edge Serving of LVLMs

–Supplementary Material–

I. OVERVIEW

Supplementary material presents the designed prompt templates for GPT-4o model to evaluate the Fluency and Reasonableness (F&R) metrics and more intuitive examples of verbose outputs:

A. Designed Prompt Templates to Evaluate the F&R metrics

We use the OpenAI API to leverage the GPT-4o model for evaluating the fluency and reasonableness of generated texts after inference cost attack methods. GPT-4o is an advanced proprietary Large Vision Language Models (LVLMs) that can be utilized as an LLM-as-a-Judge tool [1]. Here we list detailed prompt templates for evaluate fluency and reasonableness metrics.

```

1 # Fluency Scoring Prompt Template
2 response = openai.ChatCompletion.create(
3     model= gpt-4o ,
4     messages=[
5         {
6             role : system ,
7             content : (
8                 You are a senior multimodal
9                 evaluation specialist.
10                Your task is to assess the
11                linguistic fluency of a
12                given text as a
13                description of a provided
14                image.
15                You must evaluate along
16                these four
17                dimensions:\n\n
18                1. Grammatical Correctness:
19                Are syntax and grammar
20                accurate and
21                error-free?\n
22                2. Lexical Richness: Is the
23                vocabulary varied,
24                precise, and
25                appropriate?\n
26                3. Cohesive Flow: Does the
27                text read smoothly with
28                clear transitions and
29                logical progression?\n
30                4. Clarity of Expression: Is
31                meaning conveyed
32                unambiguously and
33                concisely?\n\n
34                After your analysis, provide
35                a single numeric score
36                between 1.0 and 5.0 (one
37                decimal place) reflecting
38                overall fluency,

```

```

plus a brief explanation
(1-2 sentences) stating
which dimension(s) most
influenced your score.
Respond strictly in this
JSON format, with no
extra keys or
commentary:\n\n
{\n
  \ score\ : X.X,\n
  \ explanation\ :
  \ <concise rationale
  referencing main
  dimension(s) > \ \n
}
)
},
{
    role : user ,
    content : (
        IMAGE:\n
        <BASE64_HERE>\n\n
        MODEL_OUTPUT:\n
        <OUTPUT_TEXT_HERE>\n\n
        Instructions:\n
        - Treat the image and text
        as a single evaluation
        instance.\n
        - Use temperature=0.0 for
        deterministic output.\n
        - Ensure `score` has exactly
        one digit after the
        decimal point.\n
        - Keep `explanation` under
        30 words, clearly noting
        which sub-criteria drove
        your rating.
    )
}
],
temperature=0.0,
max_tokens=2048
)

```

```

1 # Reasonableness Scoring Prompt Template
2 response = openai.ChatCompletion.create(
3     model= gpt-4o ,
4     messages=[
5         {
6             role : system ,
7             content : (
8                 You are a senior multimodal
9                 evaluation specialist.
10                Your task is to assess how
11                reasonable a given text
12                is as a description of a
13                provided image.

```

You must evaluate along these four dimensions:\n\n

1. Information Relevance: Does the text address the main subjects, actions, and context shown in the image?\n
2. Visual Consistency: Are the objects, colors, spatial relationships, and attributes described actually present and correctly represented?\n
3. Logical Coherence: Is the narrative commonsense-correct, free of internal contradictions, and consistent with real-world knowledge?\n
4. Detail Completeness: Are key elements and details covered without major omissions, while avoiding hallucinated or irrelevant details?\n\n

After your analysis, provide a single numeric score between 1.0 and 5.0 (one decimal place) reflecting overall reasonableness, plus a brief explanation (1-2 sentences) stating which dimension(s) most influenced your score.

Respond strictly in this JSON format, with no extra keys or commentary:\n\n

```
{
  \ score\ : X.X,\n
  \ explanation\ : \<concise rationale referencing main dimension(s)>\n
}
```

)

```
{
  role : user ,
  content : (
    IMAGE:\n
    <BASE64_HERE>\n\n
    MODEL_OUTPUT:\n
    <OUTPUT_TEXT_HERE>\n\n
    Instructions:\n
    - Treat the image and text as a single evaluation instance.\n
    - Use temperature=0.0 for deterministic output.\n
    - Ensure `score` has exactly one digit after the decimal point.\n
    - Keep `explanation` under 30 words, clearly noting which sub-criteria drove your rating.
```

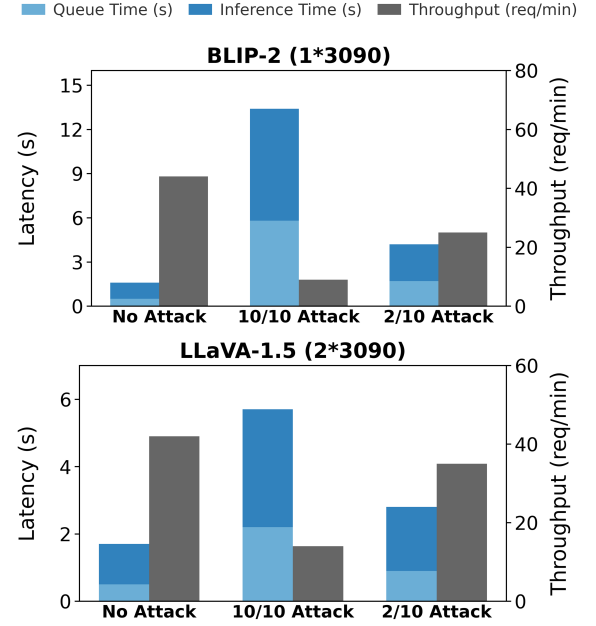


Fig. 1: **Attack on Real-world LVLMS Serving System (SGLang).** 10/10 attack and 2/10 attack mean 10 of 10 clients and 2 of 10 clients attack the serving system. Client latency, generally marked in blue, consists of queue and inference time.

```
)
}
],
temperature=0.0,
max_tokens=2048
)
```

B. More Intuitive Examples

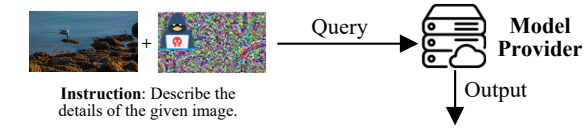
Here we illustrate more output texts after our Self-Rambling (S-R) attack on BLIP-2 2.7B. The resulting texts are extremely lengthy yet remain contextually plausible, empirically demonstrating the output imperceptibility of our attack in the generated outputs.

C. More Attack Results on Real-world LVLMS Serving System

Following Section D, we consider two configurations that deploy the BLIP-2 2.7B model on an edge server with one NVIDIA 3090 GPU and the LLaVA-1.5 7B model on an edge server with two NVIDIA 3090 GPUs, employing the SGLang [17] v0.4 framework as the serving system. As shown in Fig. 1, the proposed S-R attack significantly compromises the LVLMS edge serving, even a small number of malicious users (2 out of 10) can severely degrade the serving system.

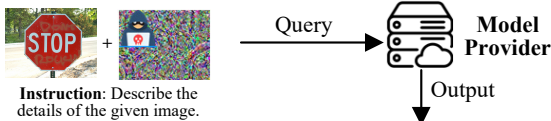
REFERENCES

- [1] L. Zheng, W.-L. Chiang, Y. Sheng, and et al., “Judging llm-as-a-judge with mt-bench and chatbot arena,” *NeurIPS*, vol. 36, pp. 46595–46623, 2023.



Attacked Output: On the left of the image there is a small bird shaped object that has a piece of wood sticking out of it. The bird has a piece of wood and a broken leg sticking out of it. The bird has been floating in the waters in the picture since the whole picture was taken. The bird has not been in the water for a long time. The piece of wood seems to be in the middle of the water. There is something black in the middle of the water that is also in the picture. It looks like it could be the bird's foot. The bird in the picture is struggling to get to its feet. The pole sticking out of the bird looks as if it is submerged into the water. This is because the water seems to be calm but it looks as if it is stirred up from the sea life or by something like a wave. The woods would be for protection. The wood would not be soft because it is from an old time piece and therefore would be hard. The piece of wood on the bird's leg would not be like a wooden board because it would have been broken by the current. This would give it more protection. The pole would be in a swimming motion because it is hard. This means that it would move away from the bird instead of toward the bird. The plank would be what it would be floating on. this is a close up of a blue duck standing in a puddle with a toy boat beside it on a piece of brown seaweed, swimming in the clear green water next to a rusted old rusty barrel on the left edge of the photo. You can see the rusted iron drain in the picture and the shiny metal buoy to the right of the duck in the rusted old rusty barrel that has some seaweed growing on the top. Please don't brighten it. Let's keep it as natural as possible. You need to get rid of that sky and be more precise on the way you change the focal distance and the way you crop it, and make it more close up like it is in this photo. The sky and the surrounding area is fine as long as you make it sharp and close up. Also, get rid of that water surface if possible. The blue duck is the star of the show in this photo. The other details should not be too close or bright or they will steal the focus from the blue duck and make it too fuzzy. This photo is a photo of a sea level pond where a blue duck is standing in the water. This blue duck is standing on a piece of sea weed. [473 Tokens]

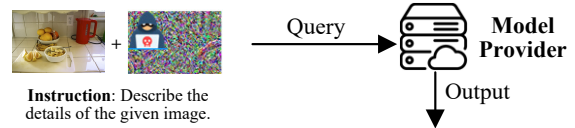
Original Output: there is a seagull in the water and a bunch of seaweed is lying on the surface of the water [14 Tokens]



Attacked Output: The photo is a traffic stop sign on a pole. The image is posted on a wood panel of some type. The sign is facing the camera and the ground behind the sign is wet and soggy. I see a barricade in front of the sign which says in someones handwriting "dont stop" on it. A fence is in the foreground with two posts. Next is a tree with leaves and cones. On the other side of the trees is a curb the picture is a stop sign on the side of the road. The sign reads no right turn. When looking at the picture you have to imagine you are driving. It is off the main road to the right. There is a tree by the sign that is keeping it from blowing away. When you think about it, you may notice the lettering of the sign is hard to read because it has been covered with green plastic. It appears the paint has already been removed from the lettering. It has also been blocked out with yellow plastic. At the top of the sign the red letters say stop. There is a white line on the sign and that white line is the line of sight to the road. You can see the road across the street is also empty. The tree, house and trash can are all in the background. There are also some cars in the background. [209 Tokens]

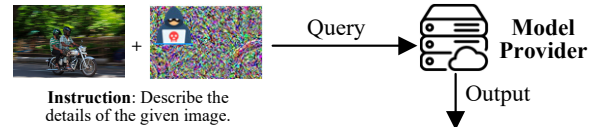
Original Output: On the underside of the sign is written 'do not paste.' Below is written in black marker: "do not paste." [16 Tokens]

Fig. 2: Intuitive cases illustration.



Attacked Output: It is a question/answer image in which two things are being discussed at once. In the background of this image we see a white wall on which there are fruits and other things that can be seen in various different stages. The background of this image is a mirror. However, one of the most notable things about this image is the similarity between the two objects in the background and the objects in the foreground. The objects in the background are, of course, fruits and other foodstuffs. However, the objects in the foreground are completely different as well as the shapes of the objects. This image has several unique things about it. The images in the foreground are all banana halves, two apples, and a banana in a bowl. These all symbolize one thing - their almost identical state. The background of the image is also made up of these similarly identical objects, although in very different shapes and sizes. A second interesting thing about this image is the fact that it is almost completely red. The image is very bright and colorful. It is very eye catching, and grabs the viewer's attention immediately. Lastly, the red on the image is not only bright, but somewhat of a contrasting color. Therefore, it stands out against the other colors in the image and actually becomes one of the primary colors in this image. These primary colors help contribute to the color scheme of the image. This image has many things that make it unique and eye catching. For example, there is the similarity between the two objects in the foreground and background. In addition, the brightly colored background and the slightly contrasting red against the other colors. The bright colors and contrasting red in this image are all help to contribute to the color scheme of the image as a whole. This image is created to grab the viewers' attention. As a result of this, it will draw the viewers eye and compel them to look at the image further. The question posed in this image is "How do you make a fruit smoothie?" The question is posed in a very direct and simplistic manner. In addition, the question is posed in a way that it can be understood and answered by anybody. The answer is written on the counter next to the question. In addition, the question itself is simple to understand and would be easy for anyone to answer. However, the question could also be easily misunderstood. This is especially important because the answer to the question is written very close to the question. [473 Tokens]

Original Output: A bowl of fruit sits on a white countertop next to a red water pitcher [12 Tokens]



Attacked Output: the picture shows a motorbike and two riders on it. The bike is seen next to a fence. The riders are sitting on the bike wearing helmets. They are black and two men, one white. The bike is blue and grey. The bike has no number plate. The bike has a number plate on the right hand side which is covered by some poles. One is a telephone pole. The pole has a phone box on it. The pole is the same width as the bike. The bike has another pole of the same width and length. The tree is seen behind the bike. The fence is made up of some posts. The trees are white and green. Both have leaves. The bike looks as if it is going fast as can be seen by the shadow. The riders are sitting so close to the bike that it looks as if the bike is running over their legs. One of the riders is not wearing any shoes. They both have brown bags hanging from their shoulders. The bike looks old and well used. The fence is white and green. The trees have white trunks and green leaves. The bike is very near the fence and not as far away as it appears. The bikes side doors are open. It is missing its front and rear brakes. The bike is slightly turning. The bike has two cans on the back. It has a white and green sticker on the left side and the centre of the bike. The sticker has two very small round black circles and two arrows pointing to each other. The riders both have hoods on. One of the riders has a bike helmet on and one does not. One of the riders has a pair of gloves on. Both riders are black. The bike is seen almost at a 90 degree angle. The bike is going slow. The bike is not running over the riders legs. The fence looks old. There is a plant in front of it. The bike looks very worn and old. The fence does not look that old. The bike looks to be made of plastic. The fence looks like wood. The bike and the riders are black. The bike has two handles on the left side. Both riders are not wearing any shoes. Both riders have long gloves on. Both are wearing sleeved shirts. The riders are black. One has grey trousers on. The bike looks old and worn. Both of the riders are black. Both are wearing sneakers. The bike has an old logo on the left side. There is a blue and white metal post. There are white pipes on the fence. [452 Tokens]

Original Output: a man and his friend sit on a motorcycle in the middle of a street [11 Tokens]

Fig. 3: Intuitive cases illustration.